

RAFFLES INSTITUTION
2013 YEAR 6 PRELIMINARY EXAMINATION

Higher 1



CHEMISTRY

8872/01

Paper 1 Multiple Choice

27 September 2013

50 minutes

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Do not open this question booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number in the spaces provided on the Answer Sheet.

There are **thirty** questions on this paper. Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider to be correct and record your choice with a soft pencil on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

This document consists of **16** printed pages.

Section A

For each question, there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct and record your choice on the OMR Answer Sheet provided.

- 1 *Use of the Data Booklet is relevant to this question.*

Which of the following substances contains the greatest amount of particles?

- A** 50 g of solid Fe_2O_3
B 1.8×10^{23} molecules of solid $\text{C}_6\text{H}_{12}\text{O}_6$
C 350 cm^3 of 2.5 mol dm^{-3} of aqueous HNO_3
D 5.0 dm^3 of gaseous H_2S at room temperature and pressure

- 2 *Use of the Data Booklet is relevant to this question.*

The mineral tellurite, TeO_2 ($M_r = 160.0$) is often used in the manufacture of optic fibres. It was found that 1.01 g of TeO_2 in an ore sample required exactly 60 cm^3 of $0.035 \text{ mol dm}^{-3}$ acidified $\text{K}_2\text{Cr}_2\text{O}_7$ for complete reaction. In this reaction, $\text{Cr}_2\text{O}_7^{2-}$ is converted into Cr^{3+} .

What is the oxidation state of Te in the final product?

- A** +2 **B** +3 **C** +5 **D** +6

- 3 The successive ionisation energies (IE) of two elements, **D** and **E**, are given below:

IE/ kJ mol^{-1}	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
D	550	1065	4138	5500	6910	8760	10230	11800
E	1140	2103	3470	4560	5760	8550	9940	18600

What is the likely formula of the compound that is formed when **D** reacts with **E**?

- A** DE_2 **B** DE **C** D_2E **D** D_2E_3

- 4 Which of the following pairs of compounds have the same shape?

- A** SO_2 and CO_2
B ICl_4^- and XeF_4
C NCl_3 and ClF_3
D BrF_5 and PCl_5

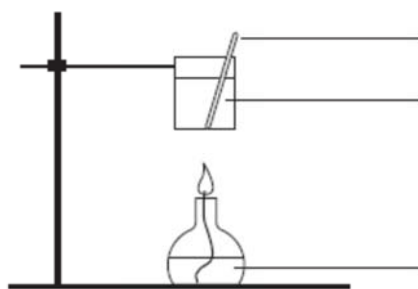
- 5 Which one of the following sequences shows the bonds arranged in order of increasing polarity?

A O – H, O – F, O – O
 B O – H, O – O, O – F
 C O – O, O – F, O – H
 D O – O, O – H, O – F

- 6 Which one of the following solids consists of atoms or molecules held only by van der Waals' forces?

A CO₂ B H₂O C SiO₂ D Cu

- 7 A student used the apparatus below to determine the standard enthalpy change of combustion of propan-1-ol, CH₃CH₂CH₂OH (*M*_r = 60.0).



thermometer
 containing 200 g water

containing
 propan-1-ol

The following results were obtained:

Mass of propan-1-ol burnt = 0.60 g
 Mass of water heated = 200 g
 Initial temperature of water = 21.0 °C

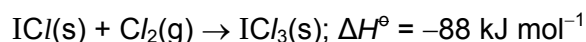
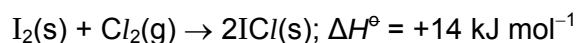
The specific heat capacity of water is 4.18 J g⁻¹ K⁻¹.

The standard enthalpy change of combustion of propan-1-ol is –2021 kJ mol⁻¹.

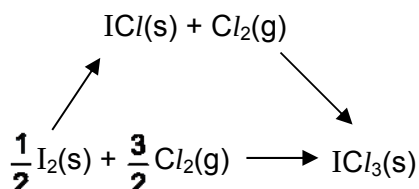
Assuming no heat loss, what would be the maximum temperature of the water?

A 24.2 °C B 29.1 °C C 45.2 °C D 48.4 °C

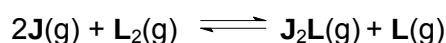
- 8 Iodine trichloride, ICl_3 , is made by reacting iodine with chlorine.



By using the above data and the following energy cycle, what is the standard enthalpy change of formation of solid iodine trichloride in kJ mol^{-1} ?



- A -60 B -74 C -81 D -162
- 9 Based on the thermochemical data below, what is the ratio of heat generated by hydrogen to that of carbon if 1 g of each substance is burned?
- Standard enthalpy change of formation of water = -285 kJ mol^{-1}
 Standard enthalpy change of formation of carbon dioxide = -394 kJ mol^{-1}
- A 0.4 B 0.7 C 3.4 D 4.3
- 10 **J** and **L**₂ react according to the following equation:



When 0.6 mole of **J** was mixed with 0.3 mole of **L**₂ at 298 K in a 2 dm³ vessel, it was found that one-third of **J** had been converted into products at equilibrium.

What is the numerical value of K_c for the reaction at 298 K?

- A 0.625 B 0.556 C 0.313 D 0.125

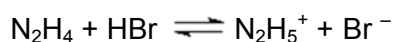
- 11 Consider the following equilibrium system:



Which of the following is correct?

	Changes made	Position of equilibrium
A	Add solid Fe_3O_4	Shifts to the right
B	Reduce the temperature	Shifts to the left
C	Reduce the concentration of $\text{CO}(\text{g})$	Shifts to the right
D	Reduce the volume of the reaction vessel	Shifts to the left

- 12 The position of equilibrium lies to the right in each of these reactions.



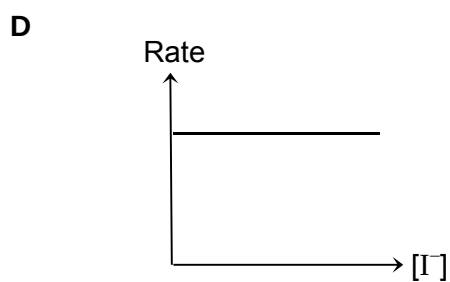
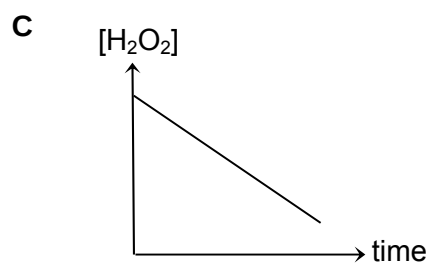
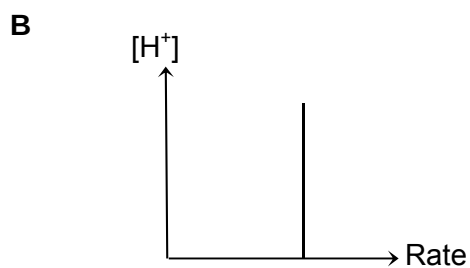
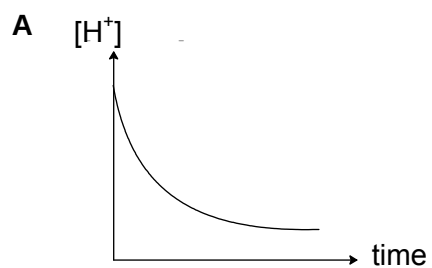
Based on this information, what is the order of acid strength?

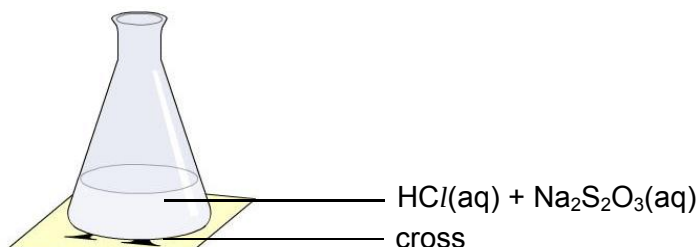
- A $\text{NH}_3 > \text{N}_2\text{H}_4 > \text{Br}^-$
- B $\text{N}_2\text{H}_5^+ > \text{HBr} > \text{NH}_4^+$
- C $\text{HBr} > \text{N}_2\text{H}_5^+ > \text{NH}_4^+$
- D $\text{N}_2\text{H}_5^+ > \text{N}_2\text{H}_4 > \text{NH}_4^+$
- 13 Which of the following 1 dm^3 acid solutions will form an acidic buffer when added to $1 \text{ dm}^3 0.100 \text{ mol dm}^{-3} \text{ NaOH}$?
- A $0.200 \text{ mol dm}^{-3} \text{ HI}$
- B $0.200 \text{ mol dm}^{-3} \text{ CH}_3\text{CO}_2\text{H}$
- C $0.050 \text{ mol dm}^{-3} (\text{COOH})_2$
- D $0.100 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$

- 14 The reaction between acidified H_2O_2 and I^- has the following rate equation:

$$\text{Rate} = k [\text{H}_2\text{O}_2] [\text{I}^-]$$

Which graph would be obtained?



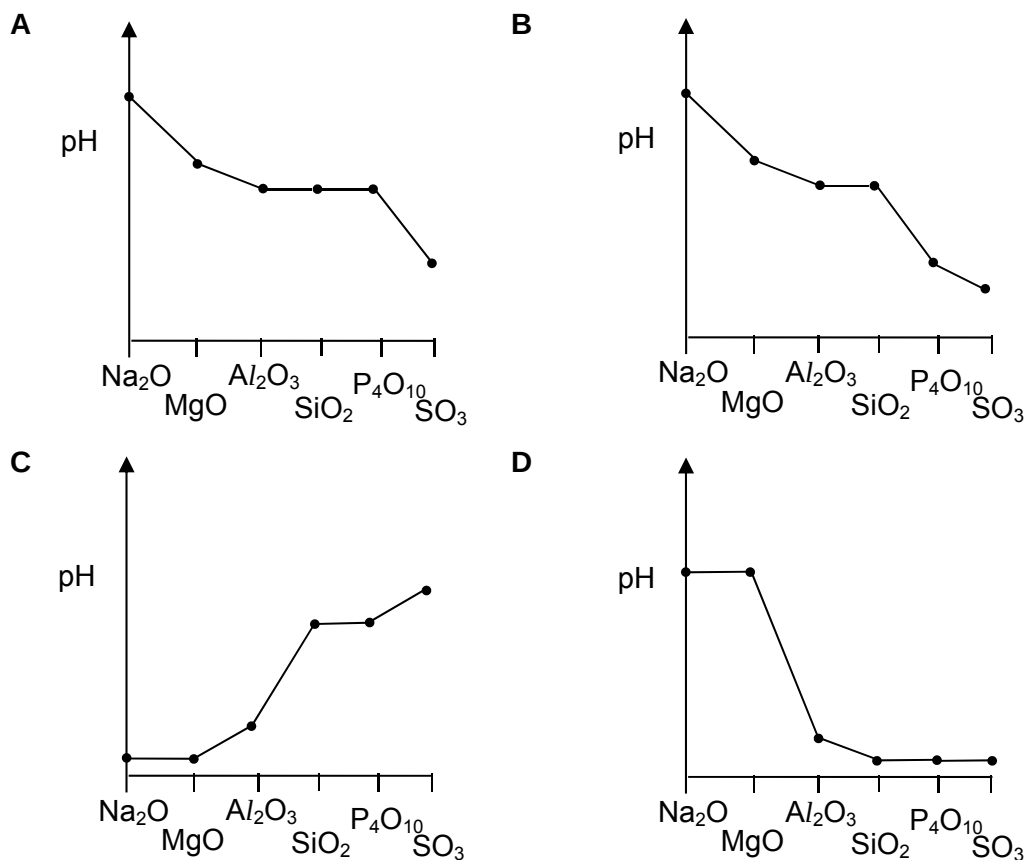


Experiment	Volume used / cm ³			t / s
	1.0 mol dm ⁻³ HCl(aq)	0.040 mol dm ⁻³ Na ₂ S ₂ O ₃ (aq)	H ₂ O(l)	
1	10.0	5.0	25.0	170
2	15.0	5.0	20.0	170
3	15.0	10.0	15.0	85
4	20.0	20.0	0.0	x

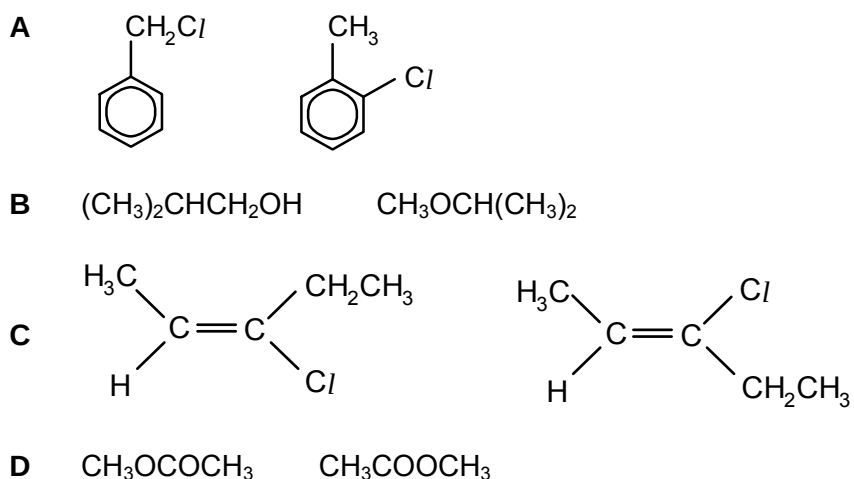
A	43	B	85
C	170	D	340

- A** Ionic radius
B Boiling point
C First ionisation energy
D Highest oxidation number in oxide

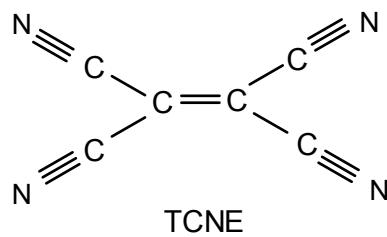
- 17 The oxides of the elements sodium to sulfur are separately added to water. Which diagram best represents the pH of the solutions produced?



- 18 Which of the following pairs of molecules are **not** isomers?



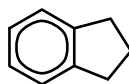
- 19 Tetracyanoethylene (TCNE) is used in the preparation of organic superconductors.



How many σ and π bonds are there in TCNE?

	σ	π
A	13	5
B	10	8
C	9	9
D	4	14

- 20** Indane is a hydrocarbon that reacts with chlorine under different conditions.

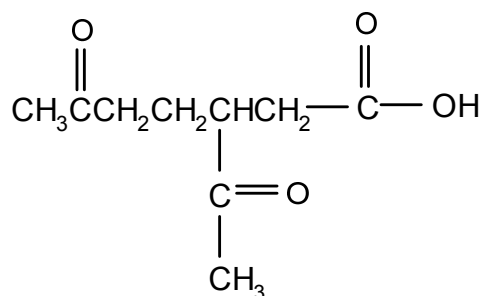


indane

Which of the following best describes the type of reaction, condition and a possible product of the reaction between indane and chlorine?

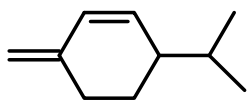
	Type of reaction	Condition	Product
A	Substitution	Ultra-violet light	
B	Substitution	Heat	
C	Addition	Fe	
D	Addition	AlCl_3	

- 21 Compound **M** reacts with excess hot acidified KMnO_4 to give the following product.

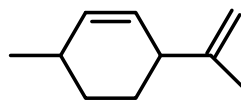


What is the structure of compound **M**?

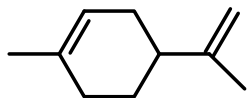
A



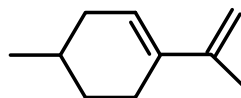
B



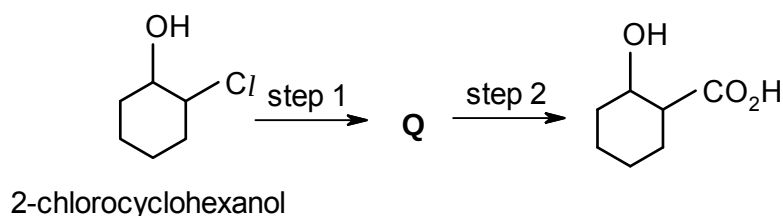
C



D



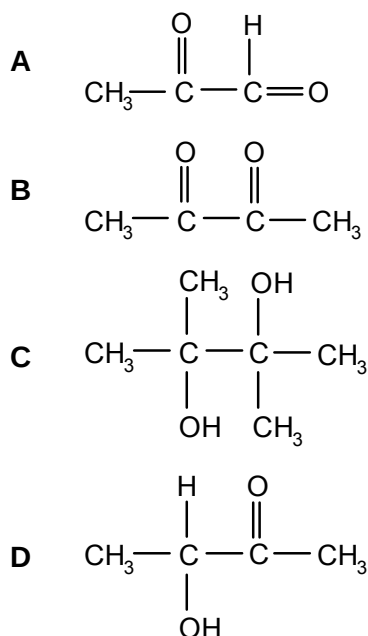
- 22 2-chlorocyclohexanol undergoes the following reaction. Which of the following are appropriate reagents and conditions for steps 1 and 2?



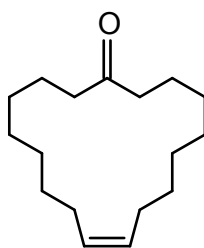
	Step 1	Step 2
A	aqueous KOH	$\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 heat
B	ethanolic KOH heat	KMnO_4 , dilute H_2SO_4 heat
C	HCN , trace KOH 10 – 20 °C	dilute H_2SO_4 heat
D	ethanolic KCN heat	dilute HCl heat

- 23** A compound has the following properties:
- ① It is unaffected by hot acidified potassium dichromate (VI);
 - ② It gives misty fumes on reacting with anhydrous thionyl chloride, SOCl_2 .

Which compound would give these results?



- 24** The naturally-occurring molecule civetone is found in a gland of the African civet cat and has been used in perfumery.

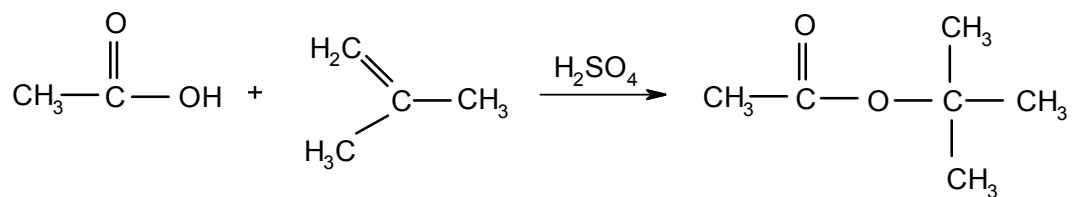


civetone

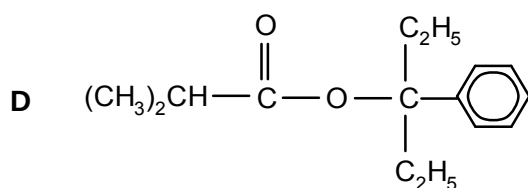
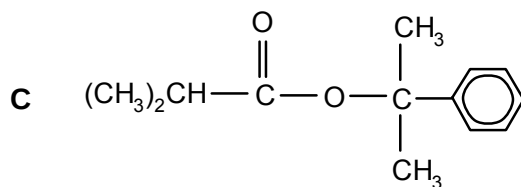
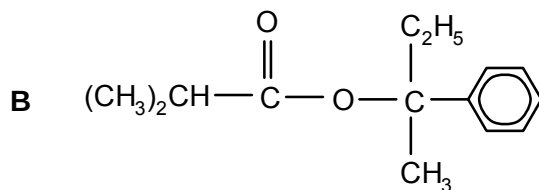
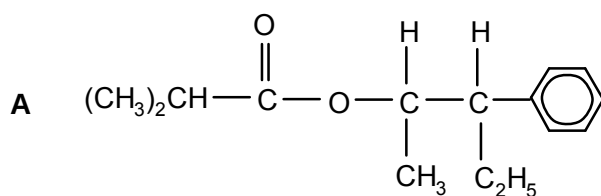
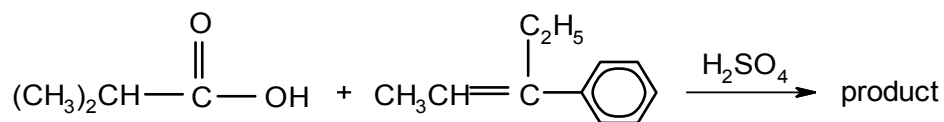
Which of the following will **not** react with civetone?

- A** Fehling's reagent
- B** hydrogen bromide
- C** 2,4-dinitrophenylhydrazine
- D** cold aqueous potassium manganate(VII)

- 25 Ethanoic acid and 2-methylpropene react in an acid-catalysed reaction as shown below.



What is the product formed in the following reaction?



Section B

For each of the following questions, one or more of the three numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

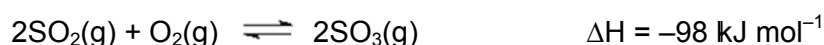
A	B	C	D
1, 2, 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** Ga, Ge, As and Se are elements in Period 4. Which of the following statements are true?

- 1** Ga^+ and Ge^{2+} are isoelectronic.
- 2** Ga^{3+} has a smaller ionic radius than As^{3-} .
- 3** Both Se and Ge have two unpaired electrons at ground state.

- 27** The following equilibrium exists between sulfur dioxide and sulfur trioxide:



Which of the following are correct?

	Changes made	K_c	Forward rate constant	Backward rate constant
1	Add catalyst	No change	Increase	Increase
2	Add more SO_2	No change	Increase	No change
3	Reduce temperature	Increase	Increase	Decrease

- 28** Which of the following statements describing the characteristics of Period 3 chlorides are correct?

- 1** In the hydrolysis of SiCl_4 , one of the products formed has a giant covalent structure.
- 2** NaCl and MgCl_2 have very high melting points whereas AlCl_3 , SiCl_4 and PCl_5 have low melting points.
- 3** An aqueous solution of AlCl_3 has a higher pH than an aqueous solution of MgCl_2 .

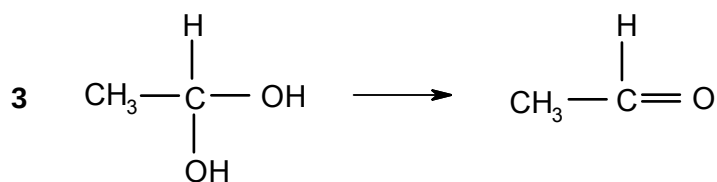
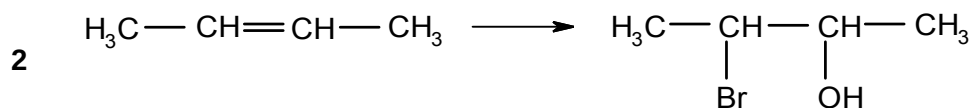
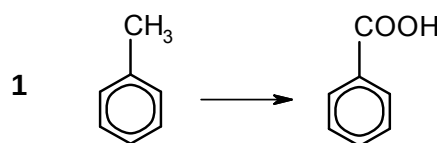
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2, 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 29** Organic reactions which involve an overall increase in the oxidation number of the carbon atoms in the molecule are considered oxidation reactions.

Which of the following conversions are oxidation reactions?

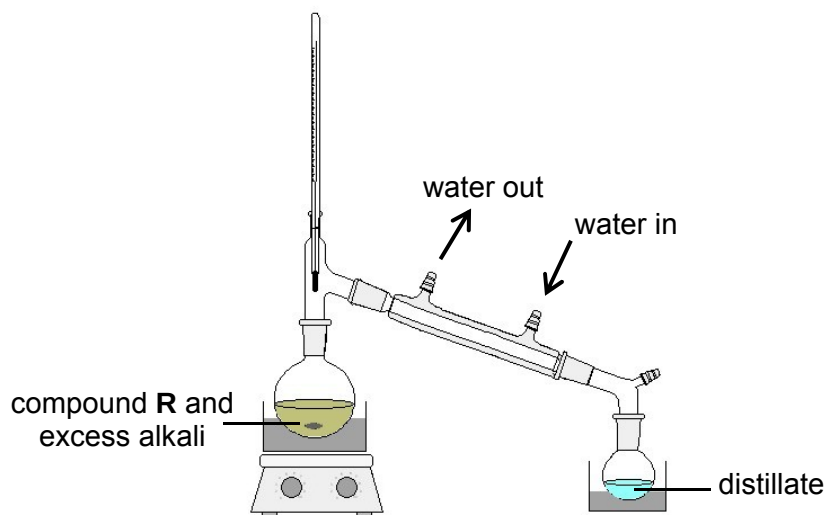


The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2, 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 30** Compound **R** is heated under reflux in the presence of excess alkali. The distillate forms a yellow precipitate with hot aqueous alkaline iodine.



Which of the following could be **R**?

- 1** ethylbenzoate
- 2** 3-iodohexane
- 3** hexanoic acid

END OF PAPER